

SHETH MVLU COLLEGE

SUBJECT:- WSN

Aim:

Sensor Mote and Base Station Communication

Study Develop a simple communication scenario between multiple sensor motes and a base station to study data aggregation and sink node operation.

Code:

SensorNode.ned

```
package sensornetwork;

simple SensorNode
{
    parameters:
        double sendInterval @unit(s);
    gates:
        output out;
}
```

BaseStation.ned

```
package sensornetwork;

simple BaseStation
{
    gates:
        input in[];
}
```

SensorNetwork.ned

```
package sensornetwork;

network SensorNetwork
{
    submodules:
        sensor[5]: SensorNode {
            parameters:
                sendInterval = 2s;
        }

        sink: BaseStation;

    connections:
        sensor[0].out --> sink.in++;
        sensor[1].out --> sink.in++;
        sensor[2].out --> sink.in++;
        sensor[3].out --> sink.in++;
        sensor[4].out --> sink.in++;
}
```

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SensorNode.cc

```
#include <omnetpp.h>
```

```
class SensorNode : public cSimpleModule
```

```
{
```

```
private:
```

```
    cMessage *timer;
```

```
protected:
```

```
    virtual void initialize()
```

```
{
```

```
    timer = new cMessage("timer");
```

```
    scheduleAt(simTime(), timer);
```

```
}
```

```
    virtual void handleMessage(cMessage *msg)
```

```
{
```

```
    if(msg == timer)
```

```
{
```

```
        cMessage *pkt = new cMessage("SensorData");
```

```
        send(pkt, "out");
```

```
        scheduleAt(simTime() + par("sendInterval").doubleValue(), timer);
```

```
    }
```

```
}
```

```
    virtual ~SensorNode()
```

```
{
```

```
        cancelAndDelete(timer);
```

```
}
```

```
};
```

```
Define_Module(SensorNode);
```

BaseStation.cc

```
#include <omnetpp.h>
```

```
class BaseStation : public cSimpleModule
```

```
{
```

```
private:
```

```
    int count;
```

```
protected:
```

```
    virtual void initialize()
```

```
{
```

```
        count = 0;
```

```
}
```

```
    virtual void handleMessage(cMessage *msg)
```

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```
{  
    count++;  
  
    EV << "Packet received. Total = "  
        << count << "\n";  
  
    delete msg;  
}  
  
virtual void finish()  
{  
    recordScalar("TotalPackets", count);  
}  
};
```

Define_Module(BaseStation);

omnetpp.ini

[General]

network = SensorNetwork

sim-time-limit = 50s

.sensor[].sendInterval = 2s

OUTPUT:

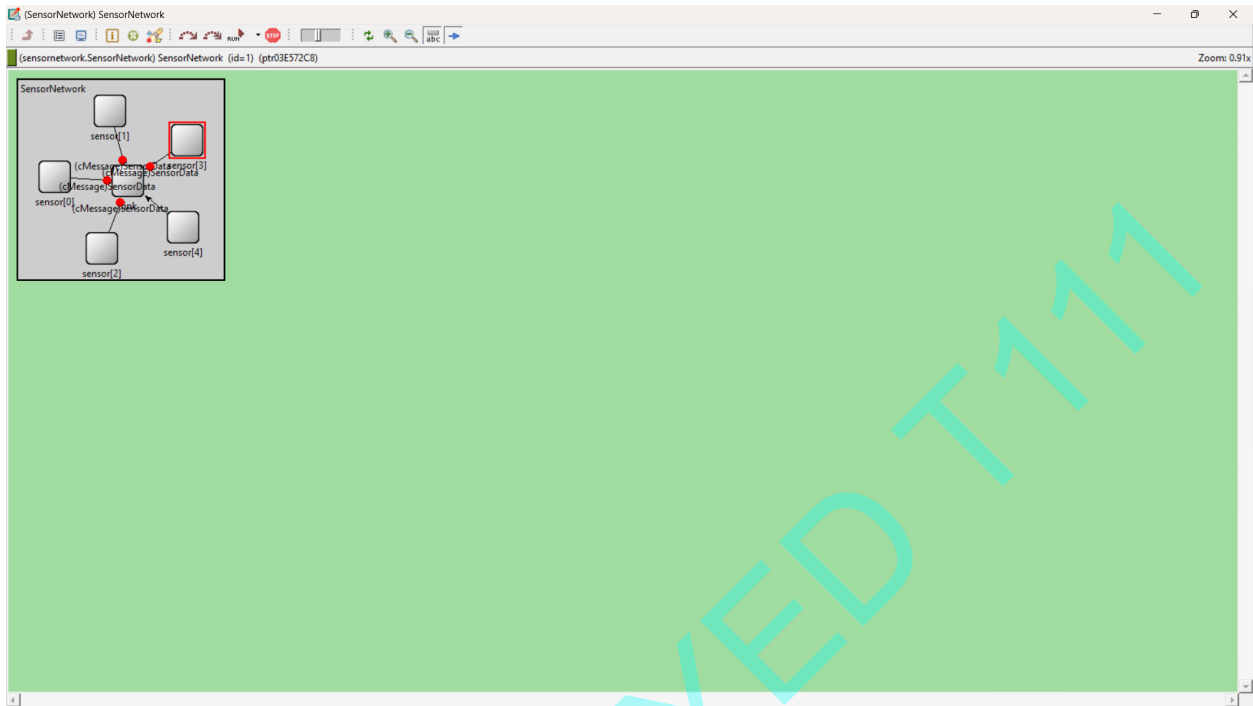
NETWORK TOPOLOGY:



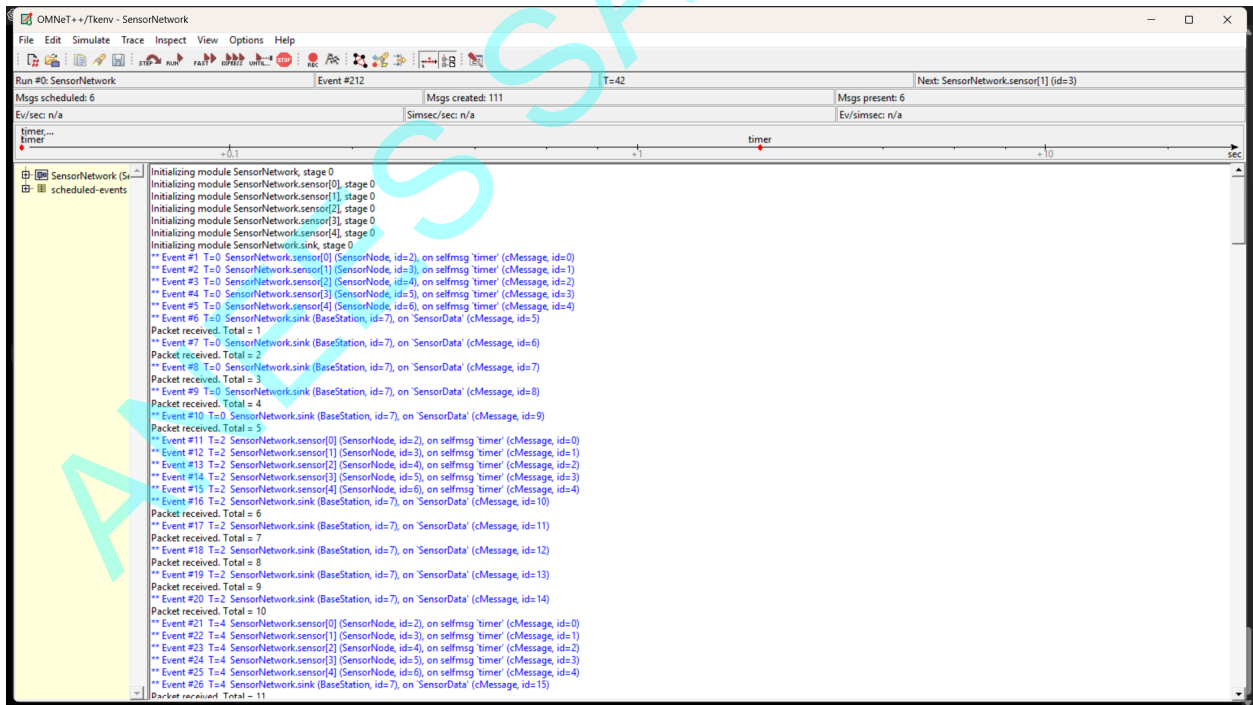
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Simulation Execution :



Event Log Packet Reception :



Result

The sensor nodes successfully transmitted data to the base station, and the sink node aggregated all received packets during the simulation.

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